

Supervised Learning Foundations Student Assessment

Book 2 Lesson 1 assessment and applied reasoning tasks

Mbaye Kebe | SRAI | Controlled review candidate | 10 September 2026

This assessment checks whether you can define a supervised learning problem, protect the prediction boundary, construct valid evidence, and distinguish a model score from an authorized decision. Complete all sections and show your reasoning.

Assessment structure

Section	Focus	Points
A	Core concepts	20
B	Mathematical reasoning	20
C	Case analysis	30
D	Notebook and professional evidence	30

Section A Core concepts

1. Define an observation, a feature, a target and prediction time in one coherent example.
2. Explain why a training score is not an estimate of future performance.
3. Distinguish validation data from test data.
4. Name four forms of leakage and give one control for each.
5. Explain why a credible baseline must use the same information boundary as the learned model.

Section B Mathematical reasoning

6. Interpret the empirical-risk objective $\hat{R}_n(f) = (1/n) \sum_i L(y_i, f(x_i))$. State what each term represents and why the result is only a proxy for expected risk.
7. Compare MAE and RMSE. Identify a decision setting in which the difference matters.
8. A seasonal baseline has RMSE 40.496 and the learned model has RMSE 30.876. Calculate the absolute and percentage improvement in RMSE.
9. Explain why ROC AUC 0.955 does not determine an operational threshold.

Section C Case analysis

A service agency predicts next-day demand. The downloaded dataset contains weather forecasts, lagged demand, region, and a field completed after the service day closes. A random split produces RMSE 2.048. A chronological split without the post-outcome field produces RMSE 30.876.

10. Identify the invalid feature category and explain the failure.
11. Select the valid performance estimate and justify your choice.
12. Propose a train, validation and test design appropriate for this task.
13. Specify two baselines, three regression diagnostics and two subgroup checks.
14. Write the conclusion that may responsibly be communicated to management.

Section D Notebook and professional evidence

15. Run the canonical notebook from the first cell and record the environment, seed and split identifiers.
16. Change one temporal cut-off and explain whether the conclusion remains stable.
17. Move the classification threshold and compare precision, recall and intervention volume.
18. List the evidence required before deployment and define one warning and one suspension trigger.
19. Prepare a one-page evidence chain linking purpose, data, model, validation, interpretation and accountability.

Submission checklist

- All quantities include units or definitions
- No test result was used for model selection
- Every feature was checked against prediction time
- Notebook outputs and written claims agree
- Limitations and human accountability are explicit